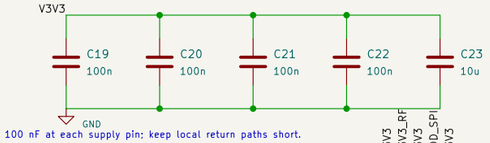
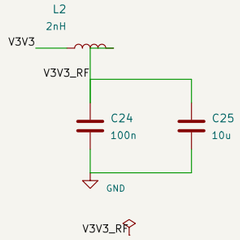
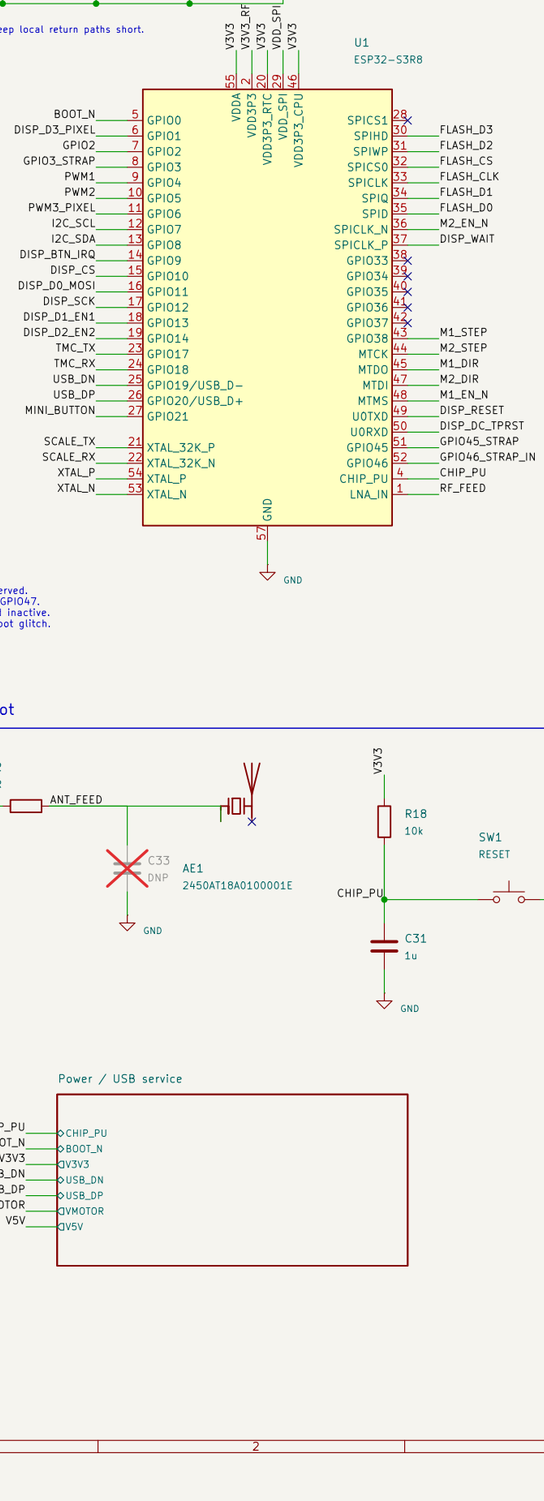
ESP32-S3R8 / one controller

V2 Pico replacement. 8 MB internal PSRAM; 16 MB external flash.



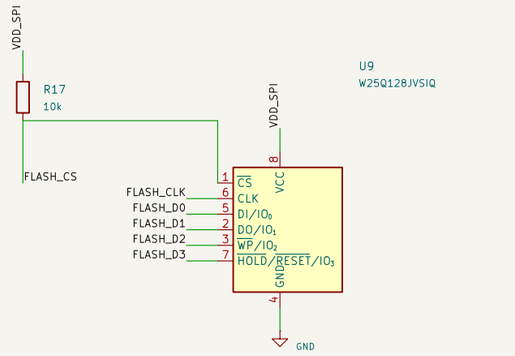
100 nF at each supply pin; keep local return paths short.

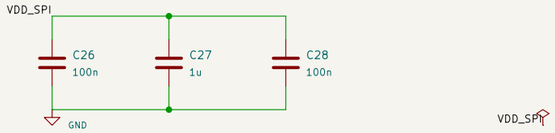




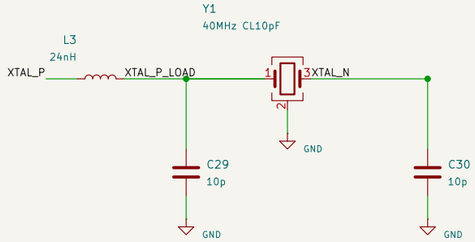
R8: pads 28, 38–42 are reserved.
Pin 36 = GPIO48; pin 37 = GPIO47.
GPIO42/48: motor EN, pulled inactive.
GPIO1/2 have a short low boot glitch.

Flash / 40 MHz clock



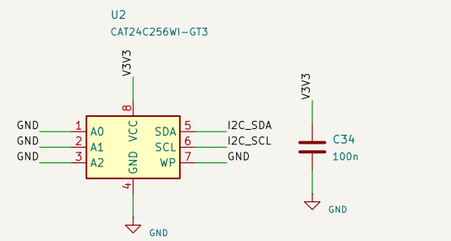


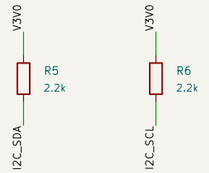
VDD_SPI supplies the 3.3 V Quad flash.
No external loads on the R8 memory pads.

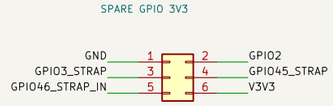


OWMC clock reference: 0402 RF/clock exceptions.
Crystal loading and antenna match checked on hardware.

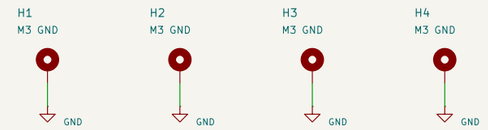
EEPROM / remaining control I/O



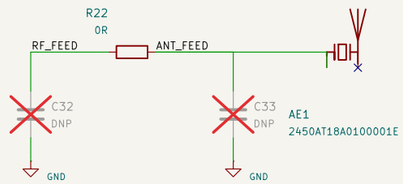


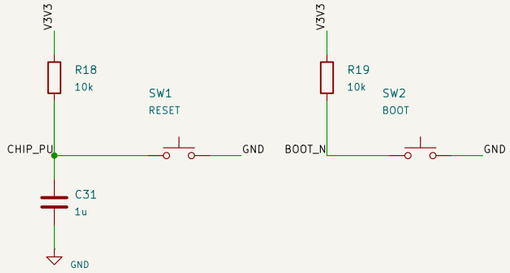


J15 is 2.54 mm; control signals only.
GPIO2: ordinary GPIO, low boot glitch.
GPIO3 / 45 / 46: boot straps; do not override the reset levels. GPIO46 reserved for inputs here.
I2C pullups use 3.0 V (child display sheet).

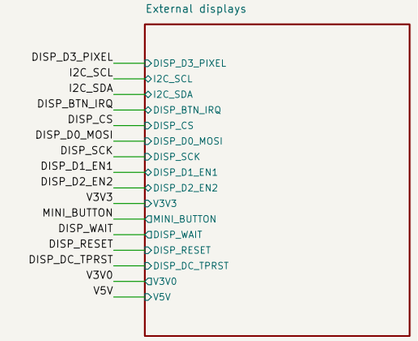


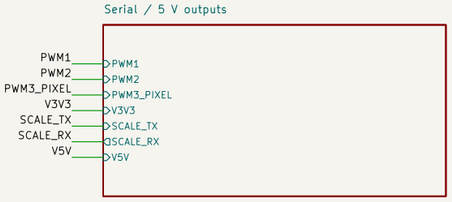
Antenna / reset / boot

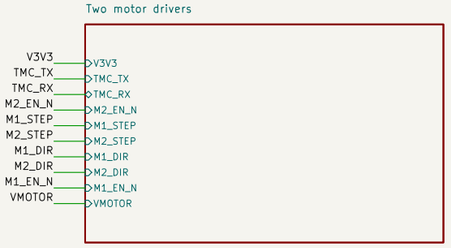












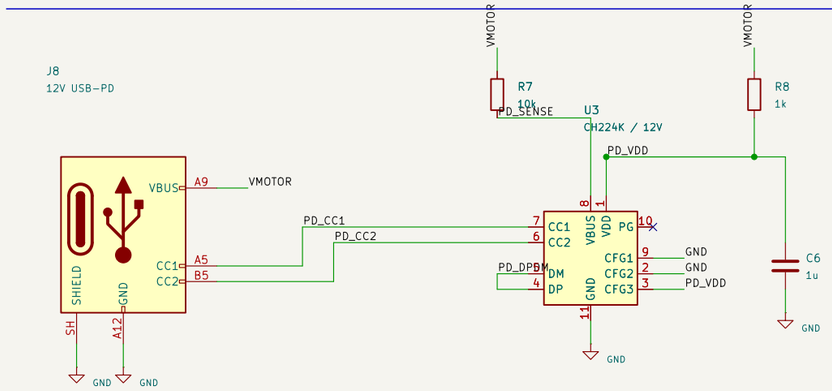
V2.1 compatible controller PCB | Review required before PCB

Sheet: /
File: motor_expansion_board_v3.kicad_sch

Title: Motor Expansion Board V3 – Controller and hierarchy

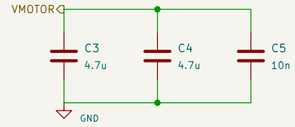
Size: A3	Date: 2026-09-13	**Rev: B1**
KiCad E.D.A. 10.0.4		Id: 1/5

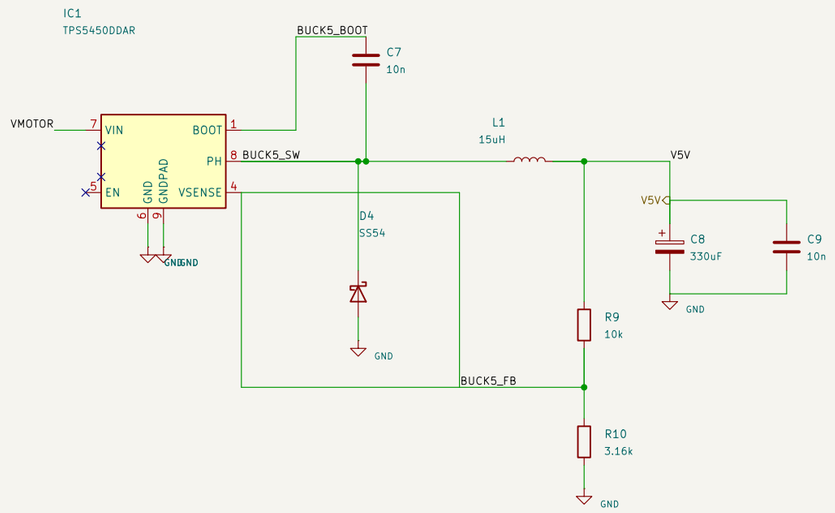
12 V USB-PD / retained V2 topology



CFG1=0, CFG2=0, CFG3=1 -> 12 V request.
Use the required 12 V PD PSU. VBUS directly feeds VM.
The PD receptacle has no USB data connection.

5 V accessories / V2 TPS5450 retained

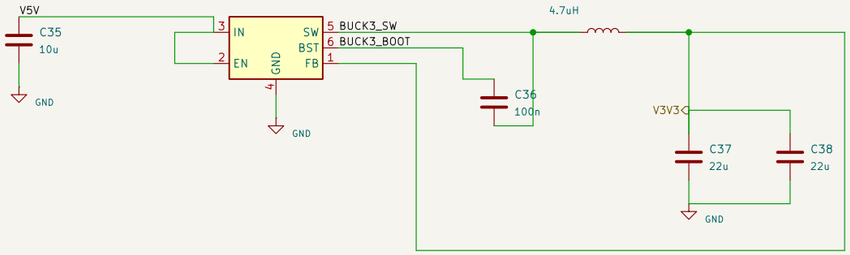




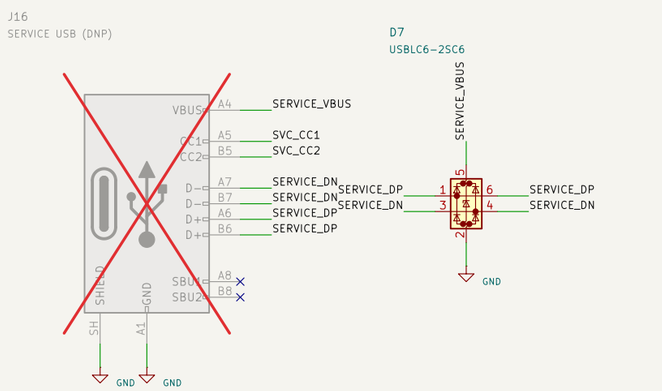
Actual IC: TPS5450DDAR; stock TPS5430 symbol uses the same pin map.
EN left open per TI. 5.085 V nominal from V2 divider.
Keep the V2 5 A class; actual board capacity needs load testing.

VMOTOR PD_VDD V5V V3V3 SERVICE_VBUS GND

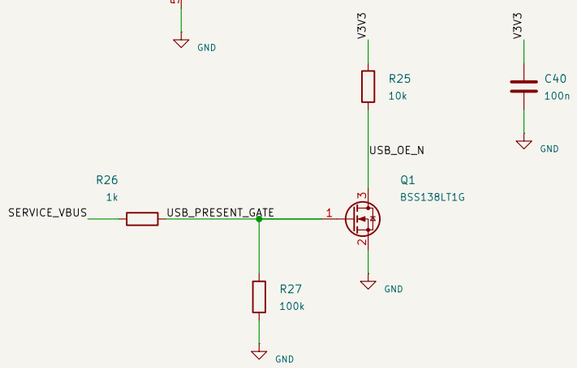
3.3 V controller supply / replaces Pico regulator



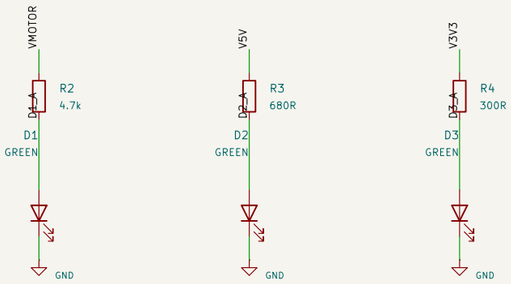
Native USB service / main PD must be on

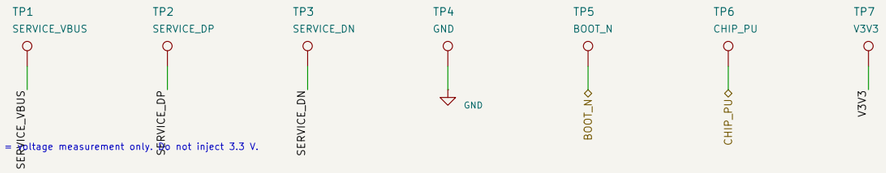






Indicators / factory contact pads





TP7 = voltage measurement only. Do not inject 3.3 V.

SERVICE_VBUS is a presence input only.
It never joins the board supply rails.
Switch loff isolates an unpowered ESP32.
J16 optional: factory pads support ROM recovery.

V2.1 compatible controller PCB | Review required before PCB

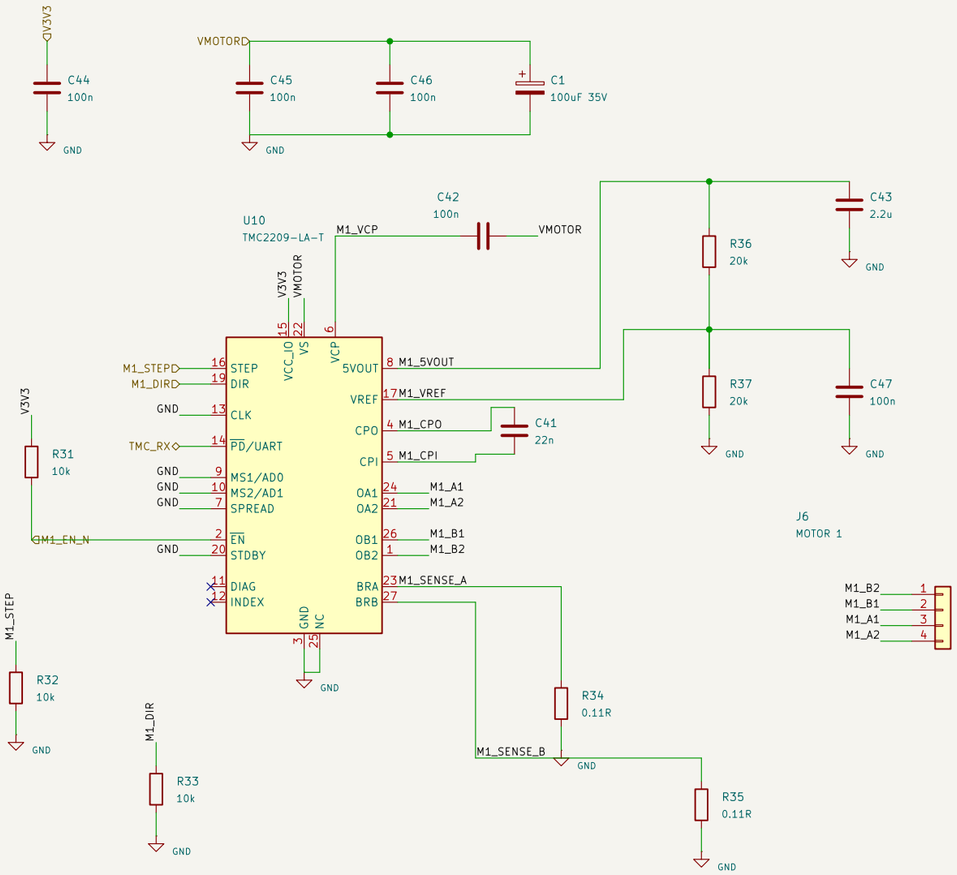
Sheet: /Power / USB service/
File: power.kicad_sch

Title: Motor Expansion Board V3 — Power and service

Size: A3	Date: 2026-09-13	**Rev: B1**
KiCad E.D.A. 10.0.4		Id: 2/5

Motor 1 / UART address 0

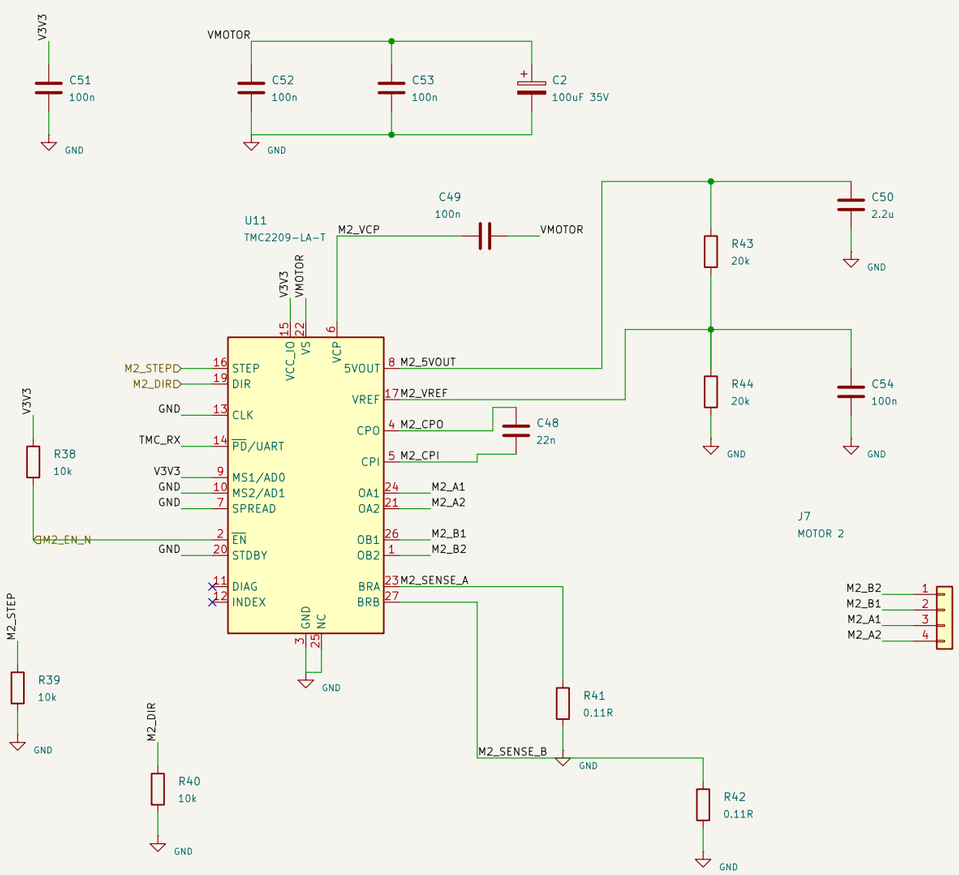
800 mA RMS default; software adjustment to 1.2–1.5 A RMS.
Upper setting is not a continuous-current rating.



Motor cable: 1=B2, 2=B1, 3=A1, 4=A2 (V2).
0.11 ohm sense resistors: 1 W / 1%; Kelvin routing.
CLK grounded: internal oscillator. SPREAD=0.
Passive heatsink: exposed pad and ground copper
follow the Fly / TMC reference implementation.

Motor 2 / UART address 1

800 mA RMS default; software adjustment to 1.2–1.5 A RMS.
Upper setting is not a continuous-current rating.



Motor cable: 1=B2, 2=B1, 3=A1, 4=A2 (V2).
0.11 ohm sense resistors: 1 W / 1%; Kelvin routing.
CLK grounded: internal oscillator. SPREAD=0.
Passive heatsink: exposed pad and ground copper
follow the Fly / TMC reference implementation.

Shared addressed UART



TX through 1k to shared RX / PDN_UART.
Initialize UART current and chopper settings before enabling.

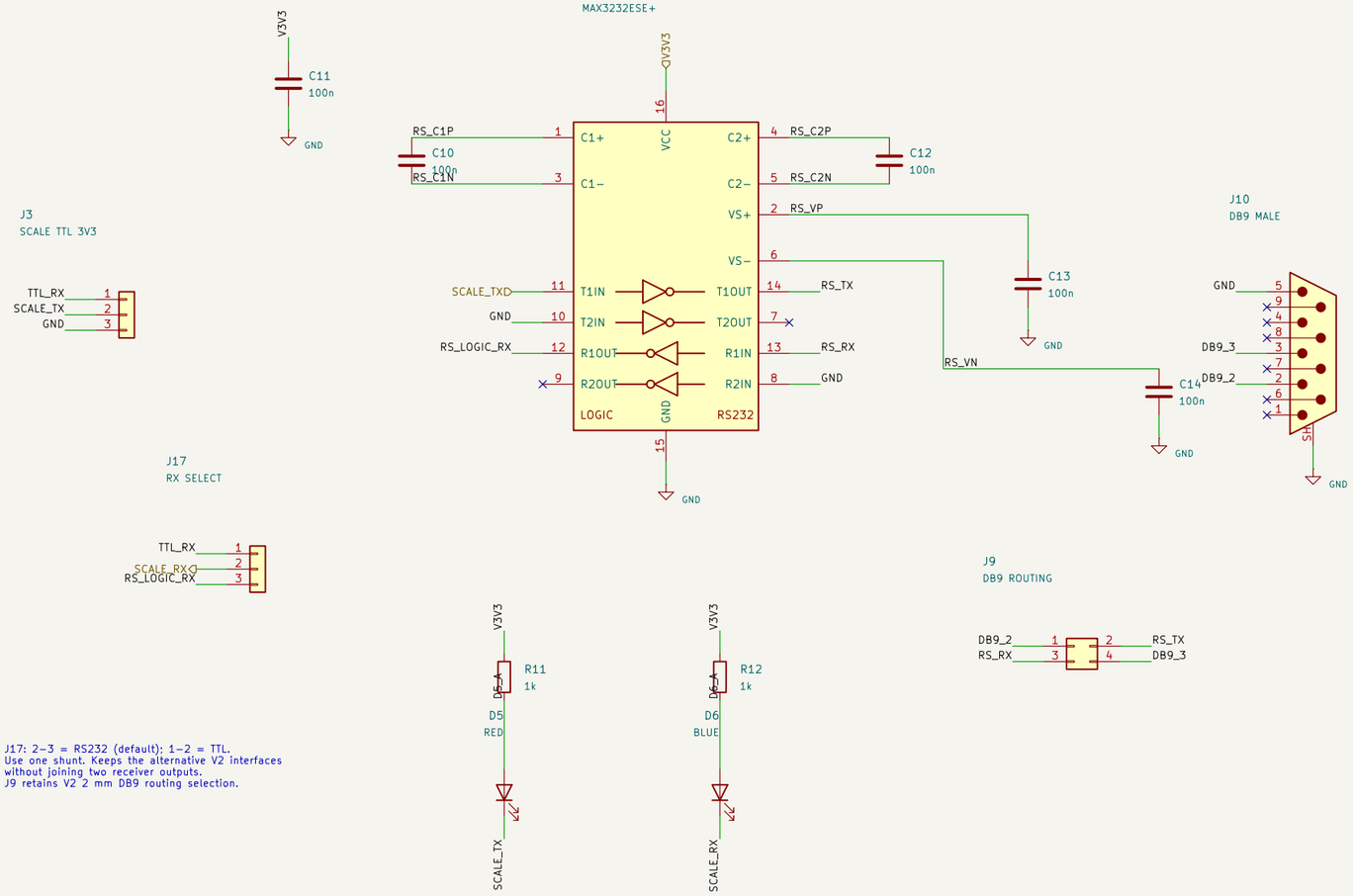
V2.1 compatible controller PCB | Review required before PCB

Sheet: /Two motor drivers/
File: motors.kicad_sch

Title: Motor Expansion Board V3 – Motor drivers

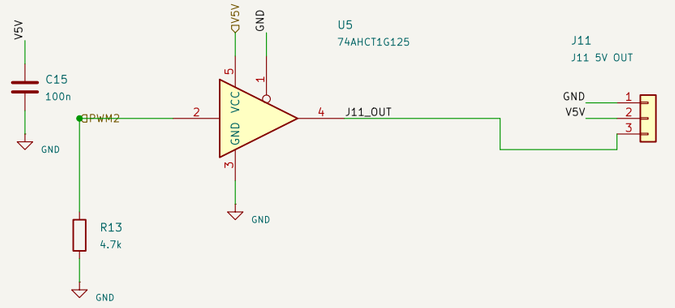
Size: A3	Date: 2026-09-13	**Rev: B1**
KiCad E.D.A. 10.0.4		Id: 3/5

Retained scale connection / TTL or RS232

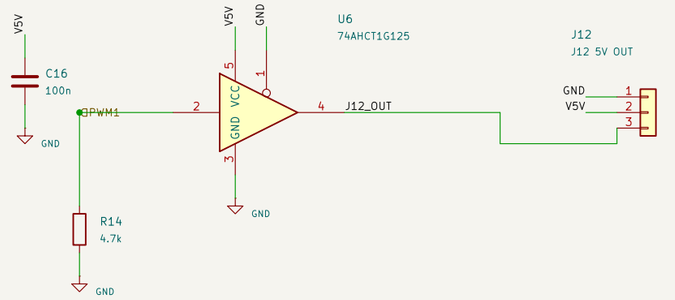


J17: 2–3 = RS232 (default); 1–2 = TTL.
Use one shunt. Keeps the alternative V2 interfaces without joining two receiver outputs.
J9 retains V2 2 mm DB9 routing selection.

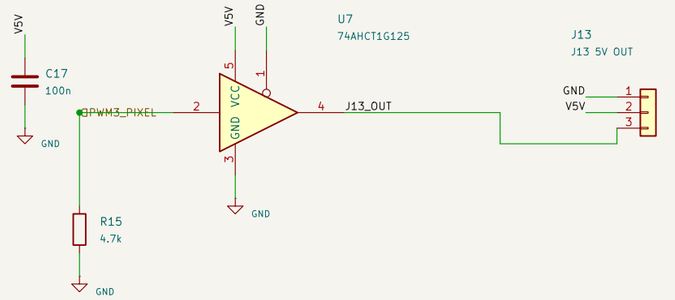
J11 – PWM2 / servo



J12 – PWM1 / servo



J13 – PWM3 / external pixels



Connector order retained: 1 GND / 2 +5V / 3 SIGNAL.
AHCT threshold guarantees a 3.3 V high with a 5 V supply.
Board load envelope: two MG90S servos + four external LEDs.

V2.1 compatible controller PCB | Review required before PCB

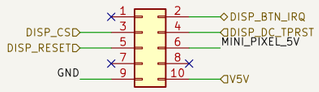
Sheet: /Serial / 5 V outputs/
File: interfaces.kicad_sch

Title: Motor Expansion Board V3 – Serial and accessory outputs

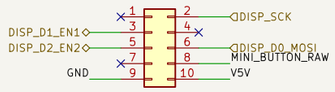
Size: A3	Date: 2026-09-13	**Rev: B1**
KiCad E.D.A. 10.0.4		Id: 4/5

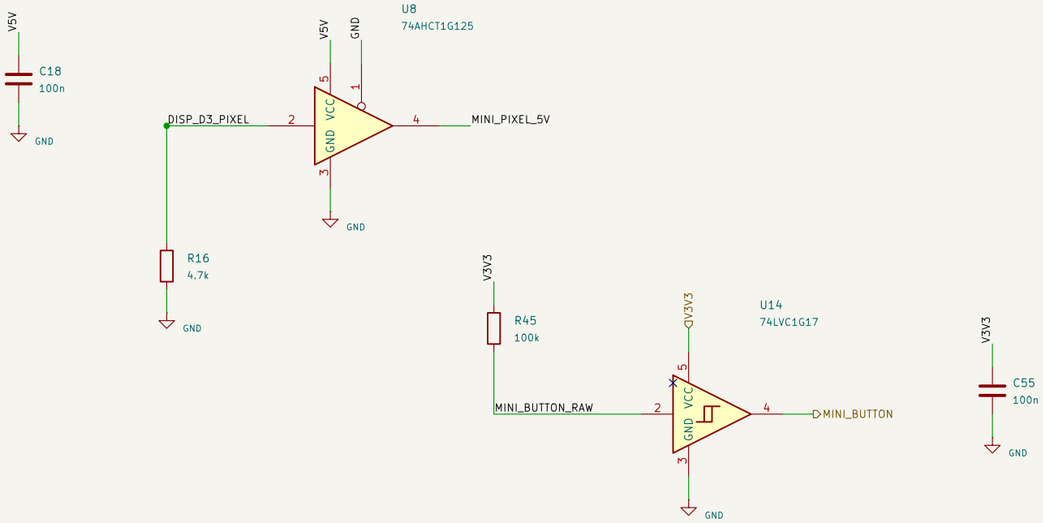
V2 Mini12864 headers / unchanged pinout

J1
EXP1 Mini12864



J2
EXP2 Mini12864

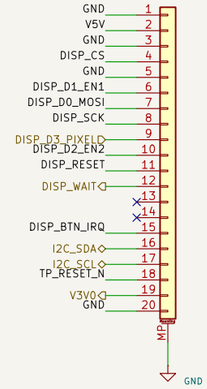


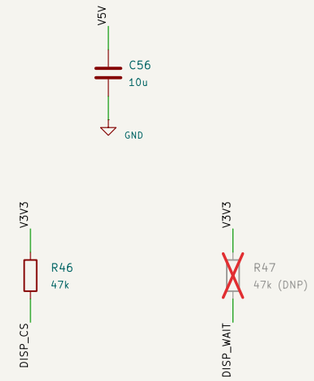


Use one display assembly at a time: power off before changing.
Mini encoder / click inputs use 3.3 V pullups in the MCU.
EXP2.8 gets a 5 V tolerant input buffer for the Mini button.
EXP2.10 stays +5 V: the supplied Mini must have no fitted
KILL link that shorts this contact when its button is pressed.

External QSPI assembly / board connector only

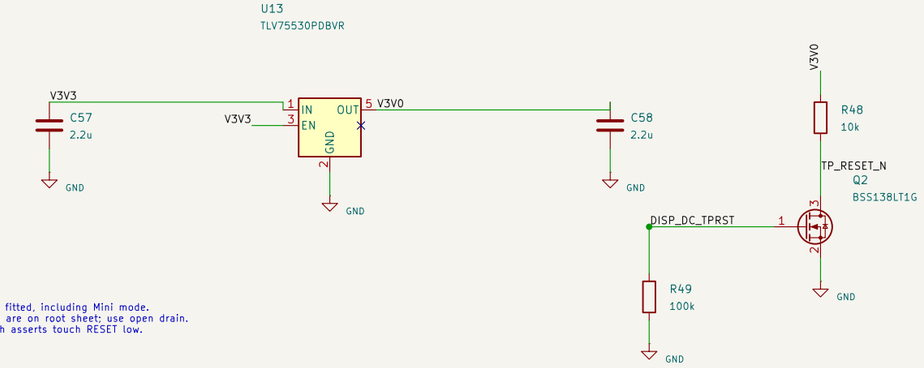
J14
TDO B1617A / 20p FFC





TDO TS040WVS02CP-B1617A: 3.95 inch, 480 x 480.
QSPI factory straps required on the supplied display.
D0..D3 / CLK / CS, RESET, WAIT and direct touch I2C.
Bridge IRQ and D/C unused for this connection.
FFC: confirm contact side, thickness and pin 1 mapping
with the supplied assembly before PCB placement.

3.0 V touch power / shared I2C pullups



LDO always fitted, including Mini mode.
I2C pullups are on root sheet: use open drain.
GPIO44 high asserts touch RESET low.

V2.1 compatible controller PCB | Review required before PCB

Sheet: /External displays/
File: display_connector.kicad_sch

Title: Motor Expansion Board V3 — External display connections

Size: A3	Date: 2026-09-13	**Rev: B1**
KiCad E.D.A. 10.0.4		Id: 5/5